Project 1

Project 1: Project Description
Supervised model for classifying vehicles using labeled geometric features. Cilj je bio naučiti model da razlikuje car, van i bus i postaviti referentnu tačku za poređenje sa unsupervised pristupom.

Project 2: Problem Description
Exploring whether vehicles naturally form meaningful groups without labels. The focus is on PCA-based data structure and KMeans clustering, and on evaluating how well the discovered clusters align with the true classes.

Project 2

Project 1

VEHICLE SILHOUETTE CLASSIFICATION

Machine Learning Project

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PROJECT WORKFLOW

01 Data Ingestion

Loading 170 vehicle instances with 18 geometric features.

02 EDA & Correlation

Visualizing relationships and statistical distributions.

03 Preprocessing

Feature scaling and train-test split (80/20).

04 Model Training

Implementing LR, SVM (RBF), and Random Forest.

05 Evaluation

Analyzing accuracy and confusion matrices.

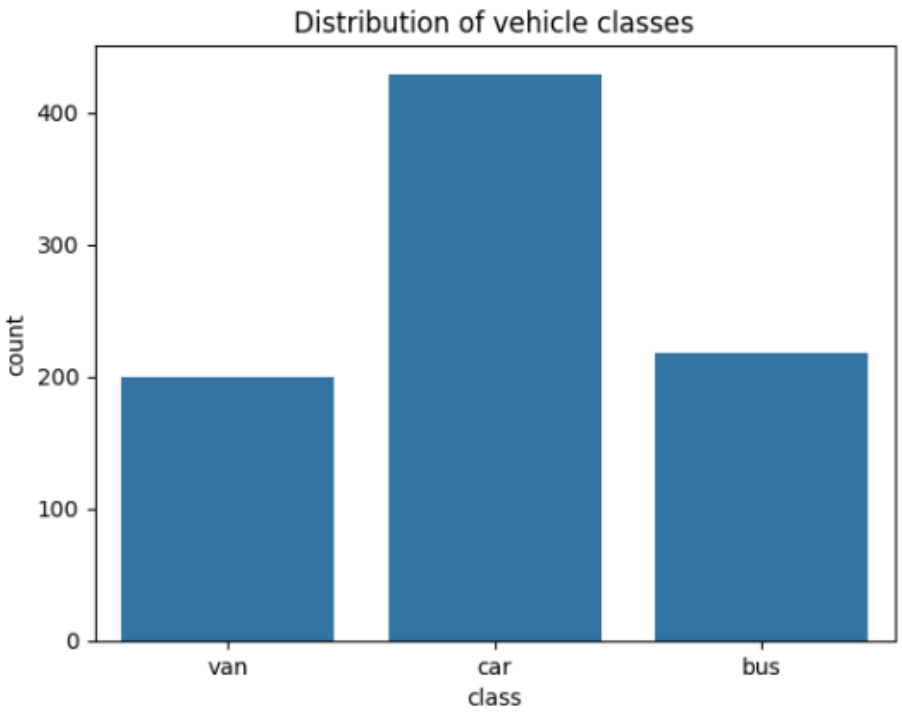
06 Comparison

Selecting the optimal model for final deployment.

A structured end-to-end Machine Learning pipeline ensuring data integrity and model excellence.

DATASET OVERVIEW

- **170 Instances** of vehicle silhouettes
- **18 Numerical** geometric features
- **3 Target Classes** : Bus, Car, Van
- **High Data Integrity** : No missing values
- **Distribution** : Slightly imbalanced classes



BASELINE MODEL: LOGISTIC REGRESSION

MODEL ACCURACY

0.93

Reference Point:
Serves as the foundational baseline for evaluation.

Stability:
Highly reliable for linear classification tasks.

Efficiency:
Low computational overhead for rapid iteration.

Constraint:
Limited performance on non-linear geometric boundaries.

📉 Logistic Regression Accuracy:
0.9352941176470588

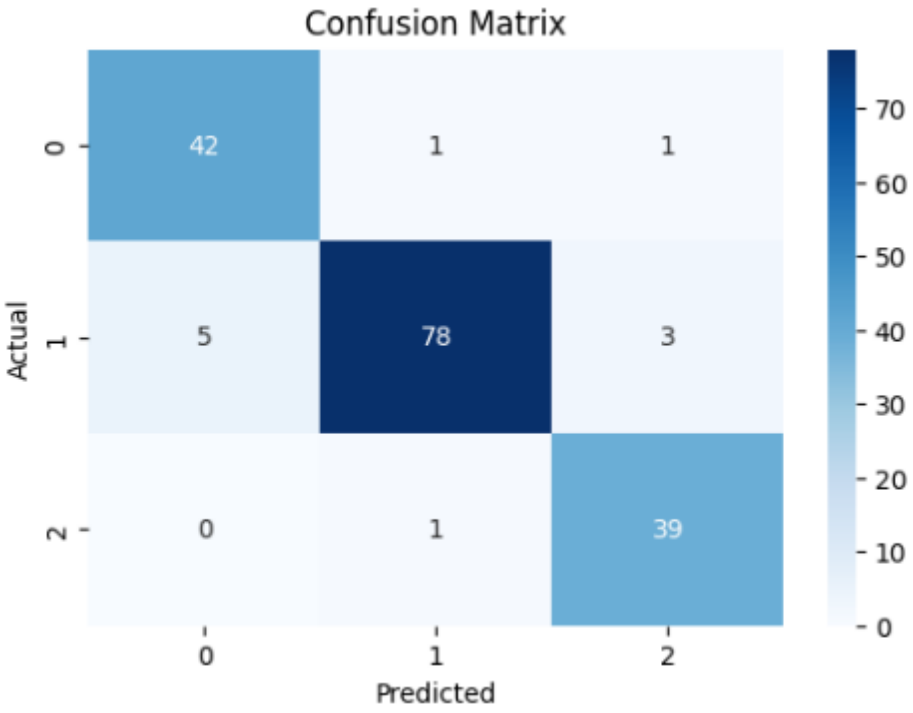
TOP PERFORMER: SVM WITH RBF KERNEL

Accuracy: 0.97
The highest performance achieved in the project.

Utilized **Radial Basis Function (RBF)** kernel for effectiv non-linear mapping.

Demonstrated exceptional precision, particularly in identifying **"Car"** silhouettes.

Proved most capable of distinguishing subtle geometric differences between classes.



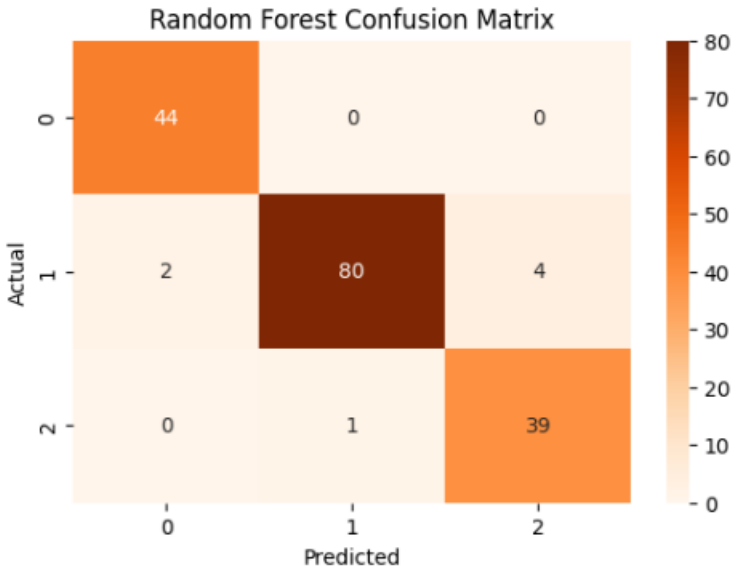
ROBUST ALTERNATIVE: RANDOM FOREST

Accuracy: 0.96
Extremely competitive performance.

Generalization
Robust model that generalizes well to unseen data.

Preprocessing
Effective performance even without extensive feature scaling.

Interpretability
Provides built-in mechanisms for feature importance analysis.



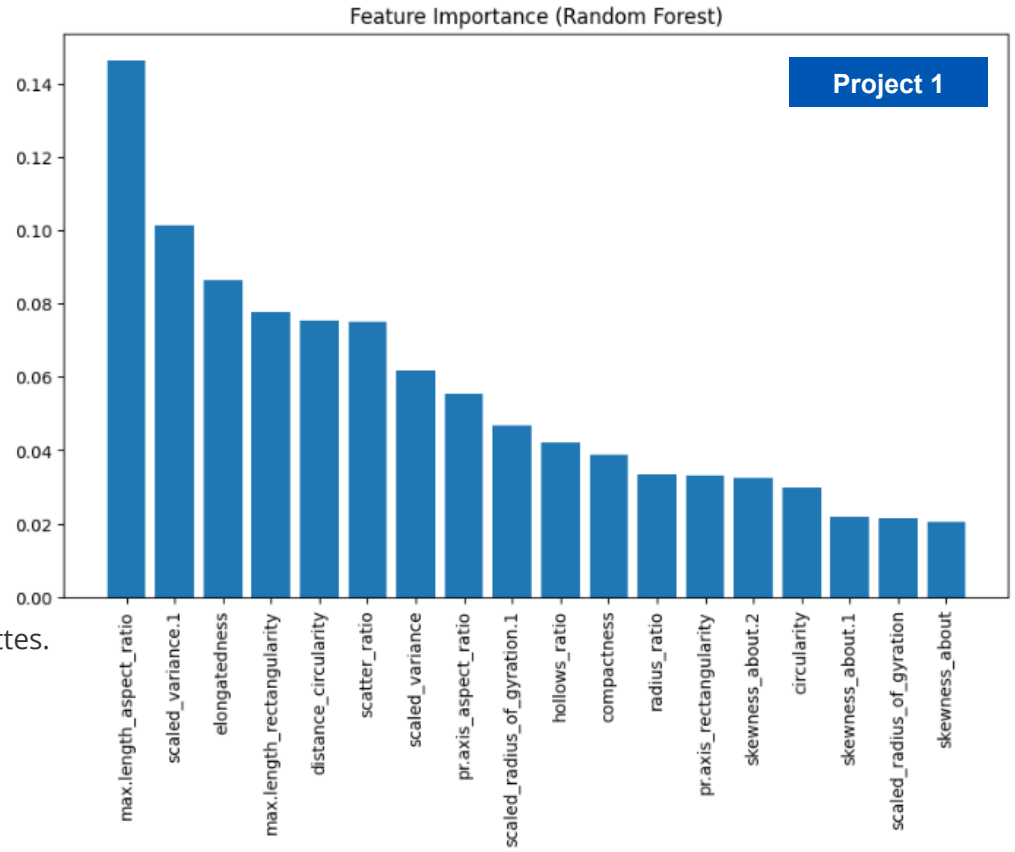
FEATURE IMPORTANCE ANALYSIS

Geometric Drivers
Identifying the most influential features for classification.

Model Interpretation
Understanding the "why" behind model decisions.

Shape Definition
Key features carry vital information about vehicle silhouettes.

Future Optimization
Insights for potential feature selection and reduction.



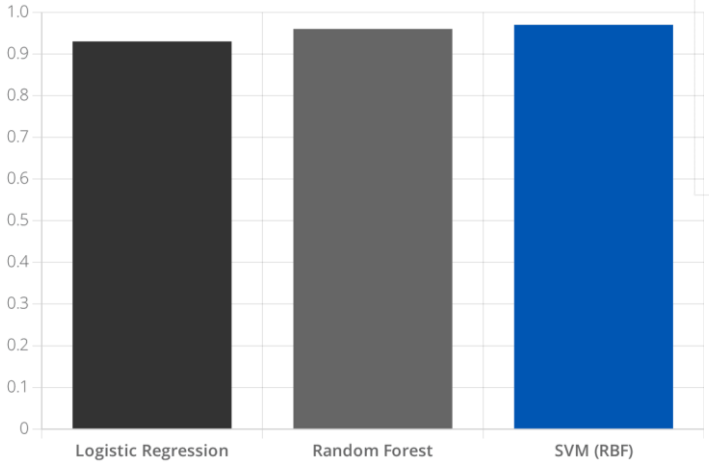
Visual representation of geometric feature significance

MODEL COMPARISON

SVM (RBF) : 0.97 Accuracy
Highest precision & performance

Random Forest : 0.96 Accuracy
Best balance of insight & stability

Logistic Regression : 0.93 Accuracy
Reliable baseline reference



The SVM model with RBF kernel outperformed others in handling non-linear geometric boundaries, while Random Forest provided valuable feature importance insights.

STRATEGIC CONCLUSIONS

Final outcomes and strategic recommendations based on the comprehensive Machine Learning evaluation.

PROJECT SUCCESS

All implemented models effectively solved the classification task with high accuracy, validating the geometric feature approach.

OPTIMAL MODEL SELECTION

SVM (RBF) is recommended for production environments due to its superior precision and handling of non-linear boundaries.

WORKFLOW MASTERY

The project successfully demonstrated a complete end-to-end ML pipeline, from data analysis to comparative evaluation.

FUTURE SCALABILITY

The framework is ready for larger datasets or real-time application deployment in vehicle identification systems.

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Project 2

PROJECT 2:
UNSUPERVISED CLUSTERING

PROJECT 2:
OVERVIEW & OBJECTIVES

Business Problem
Same classification challenge but using a different lens.

Analytical Approach
Shifting to Unsupervised Learning.

Methodology
No labels used during the training phase.

Primary Goal
Discover the natural structure and geometry within the data.

The PCA projection of true classes reveals the natural geometric structure of the dataset. Some separation is visible, but significant overlap remains, motivating the use of clustering to explore hidden patterns.

PCA DIMENSIONALITY REDUCTION

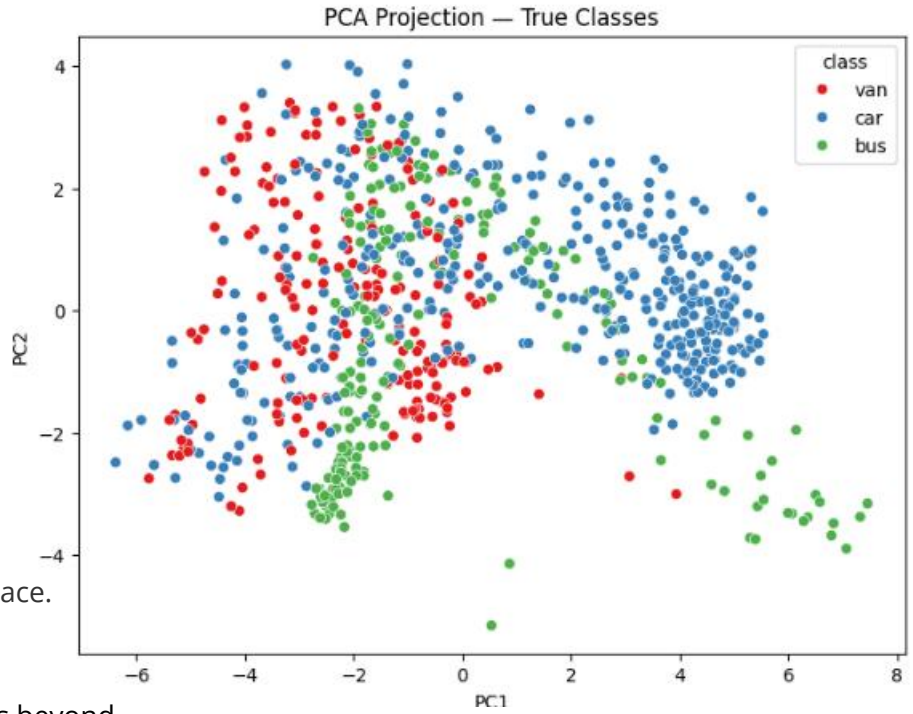
Dimensionality Reduction
Successfully reduced feature space from 18 to 2 dimensions.

Explained Variance : 68% Total
(52% PC1 + 16% PC2).

Structural Analysis
PCA reveals partial clustering but lacks full class separation.

Observation
Geometric features overlap significantly in the 2D projected space.

Motivation
This motivates the use of clustering to explore hidden patterns beyond visible structure.



The PCA projection of true classes reveals partial structure in the dataset. Some separation is visible, but significant overlap remains, indicating that the geometric features do not form fully distinct groups in 2D space.

KMEANS CLUSTERING ANALYSIS

Algorithm : KMeans Clustering implemented with **k=3**
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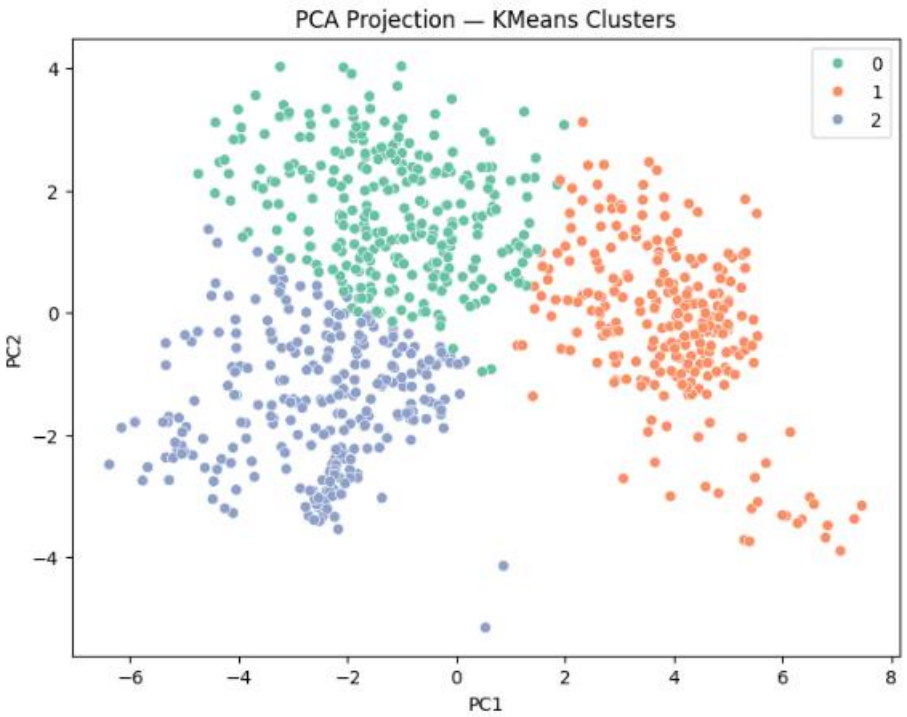
Similarity Basis
Clusters formed strictly on geometric feature proximity.

Unsupervised Nature
Zero label information provided during model training.

Resulting Structure
Discovering natural groupings based on silhouette geometry.



The KMeans clustering in PCA space reveals partial structure but significant overlap between clusters, reflecting the geometric similarity of vehicle silhouettes.

CLUSTERING EVALUATION METRICS

SILHOUETTE SCORE

0.29

Indicates weak cluster separation.

ADJUSTED RAND INDEX (ARI)

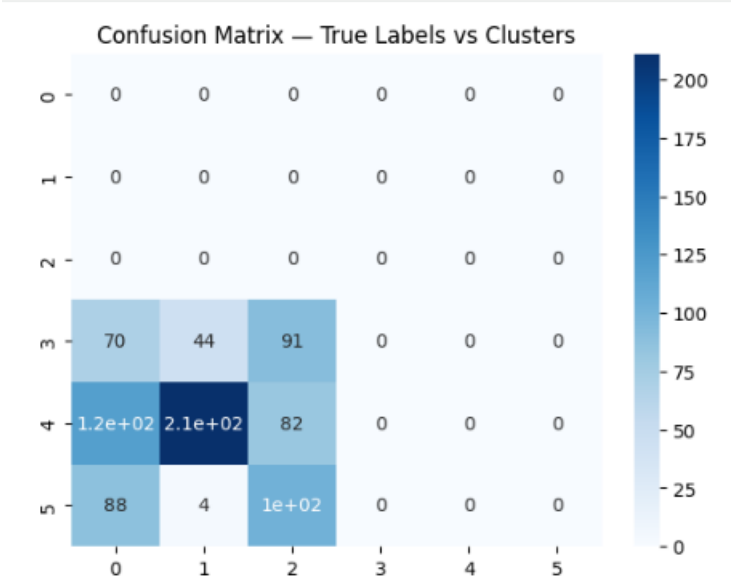
0.10

Low alignment with true classes.

NORMALIZED MUTUAL INFO (NMI)

0.11

Minimal shared information.



Confusion matrix shows that KMeans clusters do not align with the true classes. Most samples from each class are split across multiple clusters, confirming weak separation. This pattern is consistent with the low ARI and NMI scores, indicating minimal agreement between clusters and actual labels.

Project 1 ↔ Project 2

PROJECT 1 VS PROJECT 2: COMPARATIVE ANALYSIS

PROJECT 1: SUPERVISED

Accuracy: High (0.97 SVM)

Goal: Direct class prediction

Features: Labels required for training

PROJECT 2: UNSUPERVISED

Alignment: Weak (0.10 ARI)

Goal: Discovering natural structure

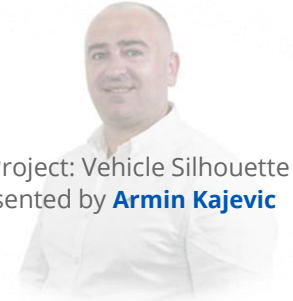
Features: Label-agnostic analysis

FINAL VERDICT

While **Supervised Learning** is essential for high-precision production tasks, **Unsupervised Learning** (PCA/KMeans) serves as a powerful exploratory tool to understand the underlying geometry and overlaps within the dataset.

QUESTIONS & ANSWERS

Thank you for your attention



Machine Learning Project: Vehicle Silhouette Classification
Presented by **Armin Kajevic**

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